

B. Tech. 1st Semester (Supplementary Mid Semester) Examination-2023

BIT Sindri, Dhanbad

Subject: Engg. Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSM 01

Answer any five questions.

1. (a) The product of any matrix with its transpose is [CO 2]
(i) symmetric (ii) skew-symmetric
(iii) Orthogonal (iv) None of these
(b) If $f_x = 3x + 2y$ and $f_y = 2x + 3y$ then $f(x,y)$ has [CO 3]
(i) minimum at (0,0) (ii) maximum at (0,0)
(iii) saddle point at (0,0) (iv) None of these
(c) The value of $\iint_R x^2 y dx dy$, where $R: \{0 \leq x \leq 2; 0 \leq y \leq 1\}$ is [CO 4]
(i) 2/3 (ii) 1 (iii) 4/3 (iv) 0
(d) The vector field $\vec{F} = y\vec{i} + x\vec{j}$ is [CO 5]
(i) irrotational (ii) solenoidal
(iii) both irrotational and solenoidal (iv) None of these
2. Determine whether (1,1,1,1), (1,2,3,2), (2,5,6,4), (2,6,8,5) form a basis of R^4 . If not find the dimension of the subspace they span. [CO 1]
3. Prove that
$$\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix} = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right).$$
 [CO 2]
4. If $f = \log r$ and $r^2 = x^2 + y^2 + z^2$, then show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{1}{r^2}$. [CO 3]
5. Prove that the product of three positive numbers of constant sum is maximum when they are equal. [CO 3]
6. Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) dx dy$ by changing to polar co-ordinates. [CO 4]
7. Find the directional derivatives of $f(x,y,z) = x^2 yz + 4xz^2$ at the point (1,2,-1) in the direction of the vector $2\vec{i} - \vec{j} - 2\vec{k}$. [CO 5]

Handwritten notes and calculations:

$r = 0$

$f = xy^2, g = x^2 yz$

$f_x = y^2, f_y = 2xy, f_z = 0$

$g_x = yz, g_y = xz, g_z = xy$

$\frac{df}{dx} = y^2, \frac{dg}{dx} = yz$

$\frac{df}{dy} = 2xy, \frac{dg}{dy} = xz$

$\frac{df}{dz} = 0, \frac{dg}{dz} = xy$

$\vec{r} = \frac{1}{\sqrt{5}}(2\vec{i} - \vec{j} - 2\vec{k})$

$\vec{r} \cdot \nabla f = \frac{1}{\sqrt{5}}(2y^2 - 2xy - 0)$

$\vec{r} \cdot \nabla g = \frac{1}{\sqrt{5}}(2yz - xz - xy)$

at (1,2,-1):

$\vec{r} \cdot \nabla f = \frac{1}{\sqrt{5}}(2(2)^2 - 2(1)(2)) = \frac{4}{\sqrt{5}}$

$\vec{r} \cdot \nabla g = \frac{1}{\sqrt{5}}(2(-1) - (1)(-1) - (1)(2)) = \frac{-3}{\sqrt{5}}$

B. Tech. 1st Semester (Mid Semester-I) Examination-2023

BIT Sindri, Dhanbad

Subject: Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSC 01

Answer any five questions.

1. (a) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ then $A^2 - 4A + 3I$ is equal to [CO 1]

(i) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(iv) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

- (b) The vectors $u=(6,2,3,4)$, $v=(0,5,-3,1)$, $w=(0,0,7,-2)$ are [CO 1]
 (i) Linearly dependent (ii) Linearly Independent
 (iii) Insufficient data (iv) None of these

- (c) The Eigen values of the matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ 8 & 4 & 0 \\ 6 & 2 & 5 \end{bmatrix}$ are [CO 1]

(i) 3,4,5

(ii) 1,1,1

(iii) 2,3,6

(iv) None

- (d) The number of asymptotes of the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ are [CO 2]

(i) 1

(ii) 2

(iii) 4

(iv) None

2. Convert the matrix $A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$ into normal form and hence find its rank. [CO 1]

3. Determine the characteristic roots and characteristics vectors of the $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. [CO 1]

4. Investigate for what values of a and b, the equations $x+2y+3z=6$; $x+3y+5z=9$, $2x+5y+az=b$ have (a) no solution (b) a unique solution (c) Infinite solution. [CO 1]

5. Find the dimension of the subspace of R_4 spanned by the set $\{(1\ 0\ 0\ 0), (0\ 1\ 0\ 0), (1\ 2\ 0\ 1), (0\ 0\ 0\ 1)\}$. Hence find its basis. [CO 1]

6. Expand $\sin x$ in powers of $x - \frac{\pi}{2}$. Hence evaluate $\sin 91^\circ$ correct to four decimal places. [CO 2]

7. Find the asymptotes of the curve $y^3 - x^2y - xy^2 + x^3 - y^2 + x^2 = 0$. [CO 2]

B. Tech. 1st Semester (Mid Semester-II) Examination-2023

BIT Sindri, Dhanbad

Subject: Engg. Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSM 01

Answer any five questions.

1. (a) If $x = r \cos \theta$, $y = r \sin \theta$ then $\frac{\partial(r, \theta)}{\partial(x, y)}$ is equal to [CO 3]

(i) r

(ii) $-r$

(iii) $\frac{1}{r}$

(iv) $-\frac{1}{r}$

- (b) If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ and $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$, then (a, b) is a maximum point of $f(x, y)$ when [CO 3]

(i) $rt - s^2 > 0$, $r > 0$ (ii) $rt - s^2 < 0$, $r > 0$ (iii) $rt - s^2 > 0$, $r < 0$ (iv) None of these

- (c) The value of $\int_1^2 \int_3^4 \int_3^4 dx dy dz$ is [CO 4]

(i) 0

(ii) 1

(iii) 2

(iv) 3

- (d) The double integral $\int_0^{\frac{\pi}{2}} \int_0^1 r \sin \theta dr d\theta$ can be evaluated as [CO 4]

(i) $\frac{1}{2}$

(ii) 1

(iii) $\frac{1}{3}$

(iv) 0

2. If $u = x + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$ then evaluate $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at $(1, -1, 0)$. [CO 3]
3. Expand $x^2y + 3y - 2$ in powers of $(x-1)$ and $(y-2)$ using Taylor's theorem. [CO 3]
4. Discuss the maxima and minima of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. [CO 3]
5. Evaluate the integral $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ using change of order of integration. [CO 4]
6. Prove that $\int_0^1 x^m (\log x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$ where n is a positive integer and $m > -1$. [CO 4]
7. Using double integration, find the area lying inside the circle $r = a \sin \theta$ and outside the cardioid $r = a(1 - \cos \theta)$. [CO 4]

BIT Sindri
Department of Chemistry
B. Tech 1st Semester
1st Mid-Semester Examination-2023
(Batch 2023-27)

Subject: Engineering Chemistry

Time : 1 hr. 30Min

Subject Code : BSC 04

Full Marks : 20

Candidates are required to give their answers in their own words as far as practicable.

Answer any **five** questions.

Q. No	Questions	Full Marks	COs
Q.1	All the Questions are multiple choices with one correct answer. Write the correct answer. (A) Which of the following is an example of QA? (a) Verification (b) Software testing (c) Validation (d) Documentation (B) The classical methods for the quantitative estimation of substances are – (a) Gravimetric methods (b) Volumetric methods (c) Both (d) None (C) An important techniques employed for the separation and purification of compounds is - (a) Chromatography (b) Conductometry (c) Potentiometry (d) Coulometry (D) A Galvanic cell converts – (a) electrical energy into heat (b) electrical energy into chemical energy (c) chemical energy into electrical energy (d) chemical energy into heat	1x4=4	CO1 & CO2
Q.2	(A) Differentiate between Quality Control and Quality Assurance in Engineering contexts. (B) Differentiate between Qualitative and Quantitative analysis in Analytical Chemistry	2x2=4	CO1
Q.3	How do you classify materials for Engineering use? Describe in detail the role of Engineering Materials in our present world.	1+3=4	CO1
Q.4	(A) Describe in detail the principle and application of IR Spectroscopy. (B) Describe in detail the principle and application of UV Spectroscopy.	2x2=4	CO1
Q.5	Define corrosion of metals. Discuss the factors influencing rate of corrosion. Explain sacrificial anodic protection of controlling corrosion.	1+2+1=4	CO2
Q.6	(A) Explain rusting of iron with the help of electrochemical theory of Corrosion. (B) Differentiate between wet and dry Corrosion.	2x2=4	CO2
Q.7	Write notes on any Four of the following: (a) Waterline Corrosion (b) Pitting Corrosion (c) Mechanical properties of an Engineering material (d) Bathochromic shift in UV-Spectroscopy (e) Laws of Dry Corrosion	1x4=4	CO1 & CO2

BIT Sindri
Department of Chemistry
B. Tech 1st Semester
2nd Mid-Semester Examination-2023
(Batch 2023-27)

Subject: Engineering Chemistry
Time : 1 hr. 30Min

Subject Code : BSC 04

Full Marks : 20

Candidates are required to give their answers in their own words as far as practicable.

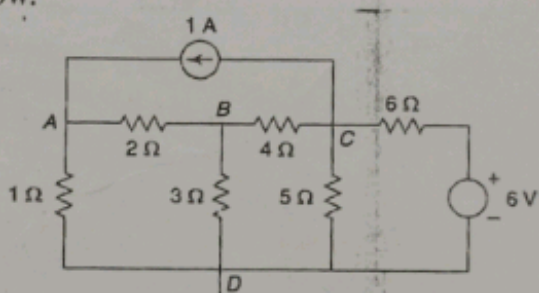
Answer any **five** questions.

Q. No	Questions	Full Marks	COs
Q.1	All the Questions are multiple choices with one correct answer. Write the correct answer. (A) Which of the electrolyte used in Li-ion battery? (a) Lead dioxide (b) Lithium-based gel (c) Sulphur dioxide (d) Cobalt (B) Which of the following material is used in solar cells? (a) Silicon (b) Barium (c) Selenium (d) Silver (C) The residual product discharged by the hydrogen-oxygen cell is - (a) Hydrogen peroxide (b) Alcohol (c) Water (d) Potassium permanganate (D) Which one of the following is an example for top-down approach? (a) Ball Milling Techniques (b) Sol-gel process (c) Both (d) none	1x4=4	CO3 & CO4
Q.2	(A) Explain Principle and construction of Lithium ion battery. (B) Distinguish between Lithium ion battery and sodium ion battery.	2x2=4	CO3
Q.3	(A) Write Working Principle and types of Fuel Cell. (B) Write Working Principle and types of solar Cell	2x2=4	CO3
Q.4	(A) Write difference between cell and Battery. (B) Write difference/Compare monocrystalline solar panels with polycrystalline solar panels.	2x2=4	CO3
Q.5	(A) Write Name of 12 Principles of Green Chemistry. (B) Write Applications of Nanomaterials in various Engineering field.	2x2=4	CO3
Q.6	(A) Define Nanomaterials. How can classify Nanomaterials into different categories depending on the number of dimensions. (B) Compare Top-down approach and Bottom up approach in the Nanotechnology with examples.	2x2=4	CO3
Q.7	Write notes on any Four of the following: (a) Polycrystalline Silicon Cell (b) Solid oxide fuel cell (c) Alkali Fuel cell (d) Green Hydrogen Technology (e) Supercapacitors	1x4 =4	CO3 & CO4



Department of Electrical Engineering
B.I.T Sindri, Dhanbad, Jharkhand- 828123
(Under the Department of Higher & Technical Education)
Govt. of Jharkhand



B.I.T Sindri, Dhanbad, Jharkhand- 828123			
Mid-Semester-3 (UG) Examinations, 2024			
Subject Code- ESBE1		Subject: Basic Electrical & Electronics Engg.	
Semester: 1 st		Department: Civil+Mining+Chemical (Section D, E & F)	
Duration: 1.5 Hrs.		Total marks: 20	
Instructions:			
(i) The marks are indicated in the right-hand margin.			
(ii) There are Seven questions in this paper.			
(iii) Attempt Five questions in all.			
Answer the following questions			
1.	(a) Convert the binary whole no. $(1101101.1011)_2$ to $(?)_{10}$ (b) Convert the Decimal no. $(45.625)_{10}$ to $(?)_2$ (c) Determine 2's Complement of binary no. 10010001 (d) convert binary no $(10110)_2$ to Gray Code	Mark 4	CO CO5
2.	Draw and explain the B-H curves for air and a magnetic material. Compare magnetic and electric circuits.	4	CO3
3.	Use node voltage analysis, determine the power in the $2\ \Omega$ and $4\ \Omega$ resistor in the network of Fig. below. 	4	CO1
4.	A voltage of 400 V is applied to a series circuit containing a resistor, an inductor and a capacitor. The respective voltages across the components are 250 V, 200 V and 180 V and the current is 5 A. Determine the phase angle of the current.	4	CO2
5.	Draw the symbols and different circuit configuration for pnp and npn BJT. What are the three regions of operation of a BJT. Explain with help of CE configuration collector characteristics.	4	CO4
6.	Draw the symbol and truth table of an NOT, OR, AND, NOR, NAND, XOR, XNOR and bubbled AND gate.	4	CO5
7.	Draw the circuit of a half wave rectifier, full wave rectifier and bridge rectifier. Also Draw its input and output waveform?	4	CO4



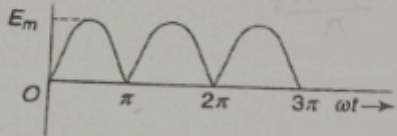
Department of Electrical Engineering

B.I.T Sindri, Dhanbad, Jharkhand- 828123

(Under the Department of Higher & Technical Education)

Govt. of Jharkhand



B.I.T Sindri, Dhanbad, Jharkhand- 828123			
Mid-Semester-2 (UG) Examinations, 2024			
Subject Code:- ESBE1		Subject: Basic Electrical & Electronics Engg.	
Semester: 1 st		Department: Civil+Mining+Chemical (Section D, E & F)	
Duration: 1.5 Hrs.		Total marks: 20	
Instructions:			
(i)	The marks are indicated in the right-hand margin.		
(ii)	There are Seven questions in this paper.		
(iii)	Attempt Five questions in all.		
Answer the following questions			
1.	Draw power Triangle and Find the rms and average value of the waveform shown in Fig. below. 	Marks 4	CO CO2
2.	A circuit consists of a coil of resistance $100\ \Omega$ and inductance $1\ \text{H}$ in series with a capacitor of capacitance $1\ \mu\text{F}$. Calculate (a) the resonant frequency, (b) current at resonant frequency and (c) voltage across each element when the supply voltage is $50\ \text{V}$.	4	CO2
3.	Define power factor as applied to ac circuits. What do you mean by active power, reactive power and apparent power?	4	CO2
4.	An ac series circuit consisting of a pure resistance of $25\ \Omega$, inductance of $0.15\ \text{H}$ and capacitance of $80\ \mu\text{F}$ is supplied from a $230\ \text{V}$, $50\ \text{Hz}$ ac source. (a) Find the impedance of the circuit, the current, the power drawn by the circuit and the power factor.	4	CO2
5.	Three similar coils each having series resistance of $20\ \Omega$ and capacitance $100\ \mu\text{F}$ are connected in star to a 3-phase, $400\ \text{V}$, $50\ \text{Hz}$ balanced supply. Find the line current, power factor, total KVA and total kW.	4	CO2
6.	Define a transformer. Also define Ideal and Practical Transformer. Discuss the principle of operation of a single-phase transformer.	4	CO3
7.	Draw the exact equivalent circuit of a transformer and describe briefly the various parameters involved in it.	4	CO3

Department of Electrical Engineering

B.I.T Sindri, Dhanbad, Jharkhand- 828123

Mid-Semester (UG) Examinations, 2023

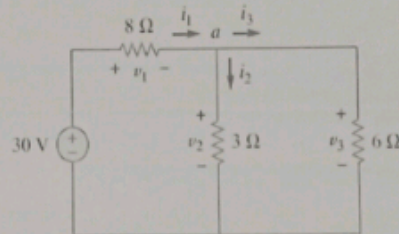
Subject Code- ESBE1	Subject: Basic Electrical & Electronics Engg.
Semester: 1 st	Department: Civil + Mining + Chemical (Section D, E & F)
Duration: 1.5 Hrs.	Total marks: 20

Instructions:

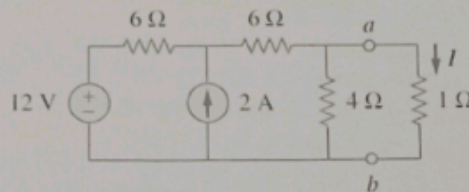
- (i) The marks are indicated in the right-hand margin.
- (ii) There are **Seven** questions in this paper.
- (iii) Attempt **Five** questions in all.

Answer the following questions

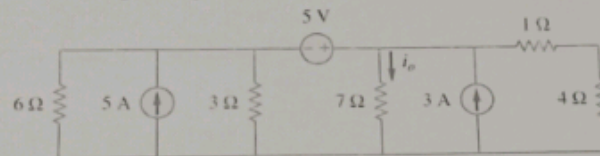
1. Find voltage V_1, V_2, V_3 & current i_1, i_2, i_3 using KCL and KVL in the ckt shown in fig.



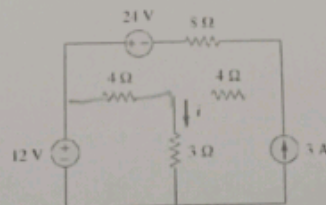
2. State Thevenin's Theorem. Using Thevenin's theorem, find the equivalent circuit to the left of the terminals (a-b) in the circuit of Fig. Then find I .



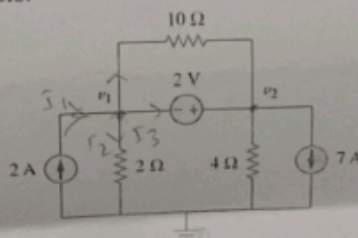
3. State and explain Kirchhoff's voltage and current laws with example. Find i_o in the circuit of fig. using source transformation.



4. State Superposition Theorem. For the circuit in fig, use superposition theorem to find i .

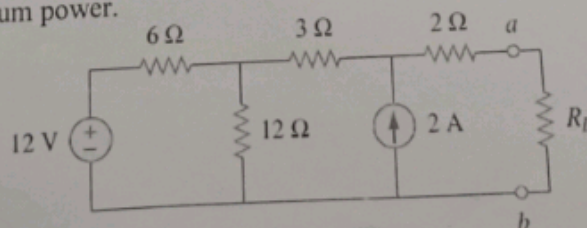


5. State Maximum power transfer Theorem with example. Find the node voltages V_1 and V_2 using nodal analysis.



$I_1 + I_2 + I_3$
 $V_1 - 2A + \frac{V_1}{2} + V_2 - 2 = 0$
 V_1

6. Find the value of R_L for maximum power transfer in the circuit of fig shown below. Find the maximum power.



7. State and explain Norton's Theorem with example. Explain different steps to find Norton's equivalent circuit.

Jharkhand University of Technology, Ranchi

BIT SINDRI, DHANBAD

B.Tech. 1st Semester

3rd SEMESTER EXAMINATION 2020

Subject: Engineering Mechanics

Time: 1:30 hrs

Full Marks: 20

Instructions:

01. Candidates are required to give the answers in their own words as far as practicable.
02. Answer any **FIVE** Questions.
03. All questions carry equal marks.

1. Figure 1 shows two vertical forces and a couple of moment 2000 N-m acting on horizontal rod which is fixed at end A.

- (I) Determine the resultant force of external forces,
- (II) Distance of resultant force from point A,
- (II) Determine an equivalent system through A.

(1 + 2 + 1 = 4 marks)

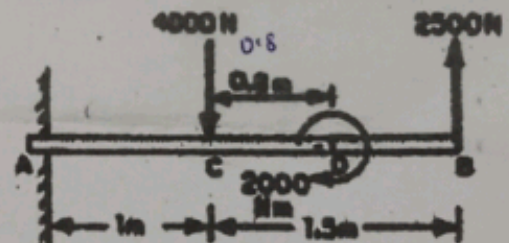


Figure-1

2. Two smooth circular cylinders, each of weight $W = 1000\text{ N}$ and radius 15 cm, are connected at their centres by a string AB of length = 40 cm and rest upon a horizontal plane, supporting above them a third cylinder of weight = 2000 N and radius 15 cm as shown in Figure 2. (I) Draw the free body diagram of each cylinder (II) Find reactions at the points of contact F and G only.

(2 + 2 = 4 marks)

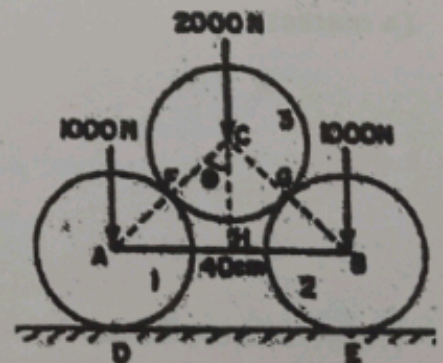


Figure-2

3. Find the expression of tensions ratio (T_1/T_2) for flat belt in terms of co-efficient of friction (μ) and angle of lap (θ). (4 marks)

4. Two blocks A and B, connected by a horizontal rod and frictionless hinges are supported on two rough planes as shown in Figure 3. The coefficients of friction are 0.3 between block A and the horizontal surface, and 0.4 between block B and the inclined surface. If the block B weighs 100 N. (i) Draw free body diagram of block A and block B (ii) what is the smallest weight of block A, that will hold the system in equilibrium?

(2 + 2 = 4 marks)

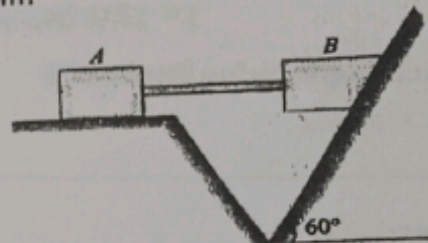


Figure-3

5. For a given Truss as shown in Figure 4. Find (i) the reactions at point C and E; and (ii) Internal force in DE using section method.

(2 + 2 = 4 marks)

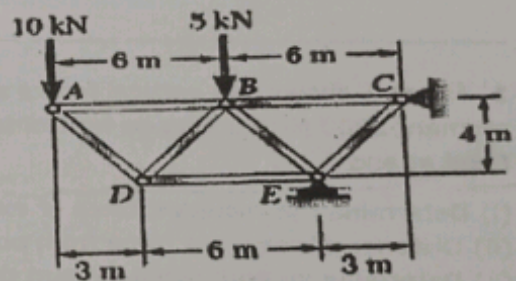


Figure-4

6. Refer Figure 5, A hammer of mass 0.5 kg hits a nail of 25 g with a velocity of 5 m/s and drives it into a fixed wooden block by 25 mm. Find the resistance offered by the wooden block.

(4 marks)

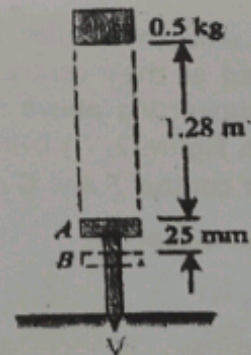


Figure-5

7. Write short notes on any two of the following:

(2 x 2 = 4 marks)

- Why couple is a Free Vector?
- Work-Energy Principle
- Assumptions in Trusses
- Impulse-Momentum Principle.

B.TECH., FIRST SEMESTER, ALL BRANCHES
FIRST MID SEMESTER EXAMINATION, 2023
ENGINEERING MECHANICS (COURSE CODE: ESEM1)

Full Marks: 20

Time: 1 Hour 30 Minutes

Roll No.

2

3

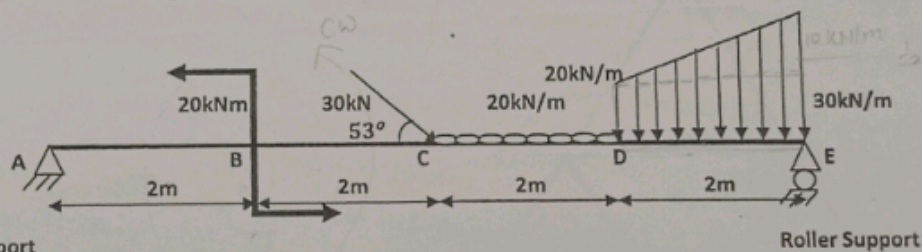
Important Instructions:

1. ALL THE QUESTIONS CARRY EQUAL MARKS.
2. ANSWER ANY FOUR QUESTIONS.
3. The full marks corresponding to a question is indicated in right braces.
4. Start answering a new question on a new page.
5. Answer all the subparts of a question at one place in continuation.
6. Assume any missing data or values suitably.
7. Candidates are required to write their answers in their own words as far as practicable.
8. Write your Roll No. on the question paper in the space provided.
9. Do not write anything else on your question paper.
10. Use of a non-programmable calculator is permitted.
11. Use of any Log Tables, Slide Rules, Data Books etc. is not permitted.

1. Three mutually perpendicular forces $\vec{F}_1 = (4\vec{i} + 5\vec{j})$ kN, $\vec{F}_2 = (5\vec{j} + 3\vec{k})$ kN, $\vec{F}_3 = (4\vec{i} + 3\vec{k})$ kN are acting on a large cubical block at the points $A \equiv (3m, 0m, 0m)$, $B \equiv (0m, 4m, 0m)$ and $C \equiv (0m, 0m, 5m)$ respectively.
 - (i.) Find the net resultant force acting on the cube.
 - (ii.) Find the net resultant moment acting on the cube about the point $O \equiv (0m, 0m, 0m)$.
 - (iii.) Find the moment about x-axis, y-axis and z-axis.
 - (iv.) Find the simplest equivalent system.
 - (v.) Find the intersection of the axis of simplest equivalent system with xy, yz and xz planes.
 - (vi.) Define the pitch of a wrench. Find the pitch of the wrench, if formed.
 - (vii.) State when a wrench is said to be right-handed or left-handed. State whether the wrench is right-handed or left-handed, if formed.

[0.50 + 0.75 X 6 = 5]

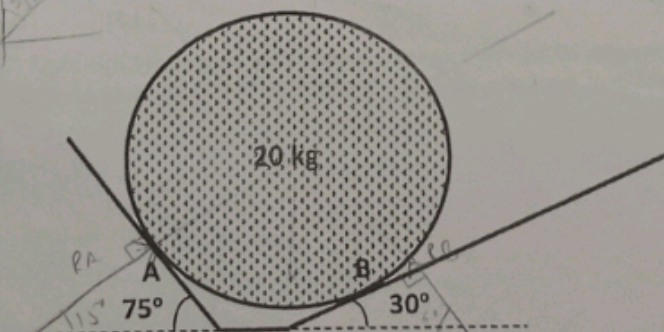
2. (i.) What is moment of a couple? Why it is called a free vector? [0.5]
- (ii.) A system of forces and couple acting on a simply supported beam is shown. Find the reaction forces acting at its supports A and E. [4.5]



Pin Support

Roller Support

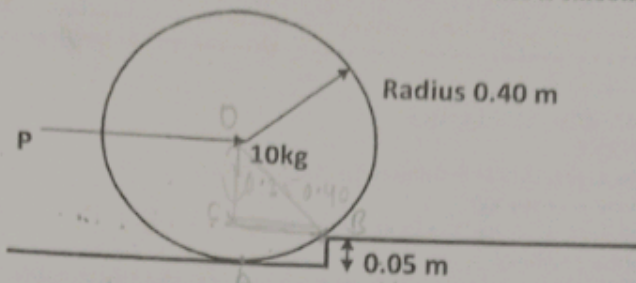
3. (i.) What are the body forces and surface forces? How they are represented in a free body diagram? [1]
- (ii.) A 20 kg homogeneous smooth sphere rests on the two inclines as shown in figure. Determine the contact forces at the points A and B.



[4]

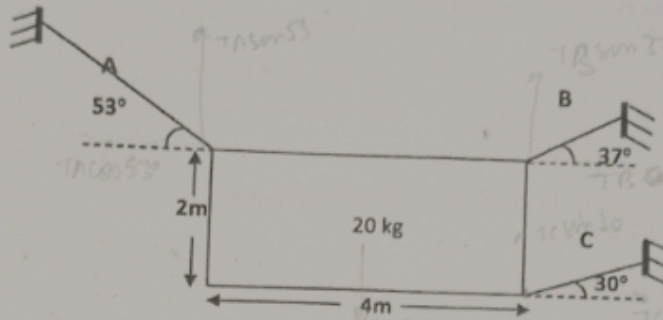
4. The door of a window is 2 m high, 1 m wide and weighs 10 kg. The door is supported by two very small hinges situated at a side at a distance of 0.5 m from the top and bottom ends. If the magnitudes of the forces exerted by the two hinges on the doors are equal, then find the reactions at the two hinges. [5]

5. Determine the minimum force P necessary to push a roller having radius 0.40 m and weight 10 kg over an obstruction having height 0.05 m as shown in figure. Force P is applied at the center of the roller horizontally. The roller surface is smooth.



[5]

6. A rectangular uniform thin slab having mass 20 kg and dimensions $4\text{ m} \times 2\text{ m}$ is supported by three light cables A, B and C tied to three fixed supports as shown in figure. Find the tension in the three cables. All the cables and slab lie in the same vertical plane.

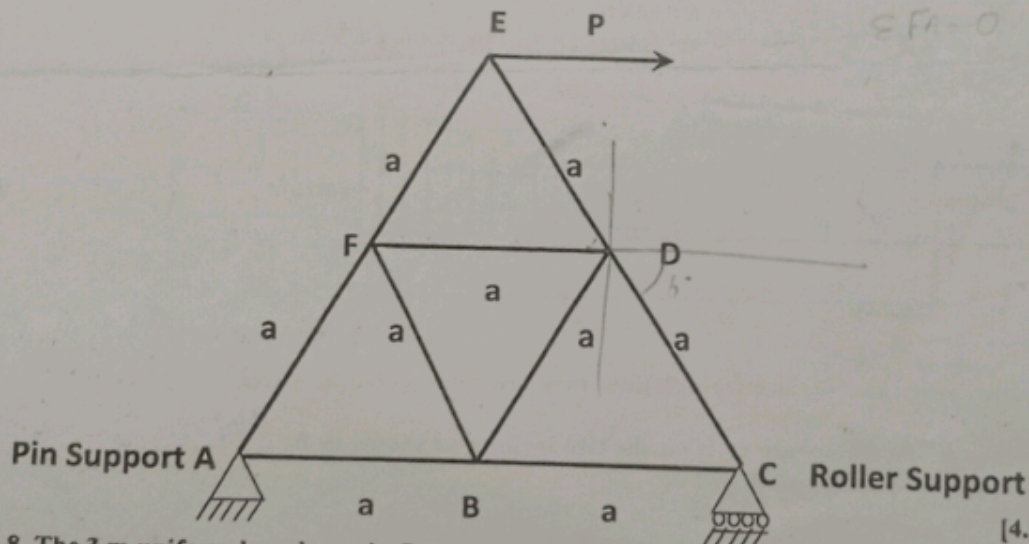


[5]

7. (i.) Write the assumptions of a plane truss.

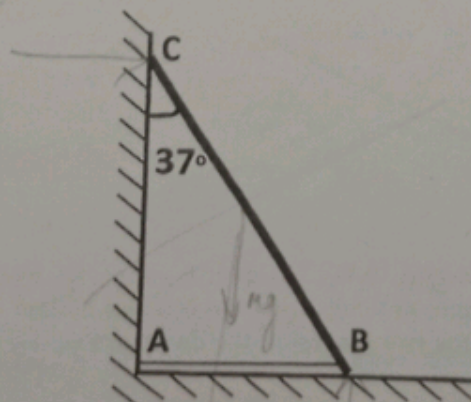
[0.5]

- (ii.) The equiangular truss is loaded and supported as shown. The length of each member is ' a '. A horizontal load P is acting at E in right direction. Determine the forces in all the members in terms of the horizontal load P .



[4.5]

8. The 3 m uniform bar shown in figure weighs 20 kg rests against a wall with a support by string at bottom. The floor and the vertical wall are smooth. Determine the tension in the string AB.



[5]

THE END

BIT SINDRI DHANBAD (JHARKHAND) – 828 123
Mid Semester Exam - II (ALL BRANCH - 2023)
1th Semester,
SUBJECT: ENGINEERING MECHANICS

Time Allowed: 90 Minutes

Full Marks: 20

INSTRUCTIONS

1. There are 06 questions answer any 04 questions. Each question carries equal marks.
2. Notations are their usual meaning.
3. Assume suitable data for any missing data.

- Q.1** a) If a particle moving with velocity, v on a curved path of radius of curvature is ρ then write the scalar components of the acceleration and show its directions. [2]
- b) The position of a particle which moves along a straight line is defined by the relation, $x = t^3 - 6t^2 - 15t + 40$, where x is expressed in meters and t in seconds. Determine:
i) the time at which the velocity will be zero,
ii) the distance traveled by the particle at that time,
iii) the distance traveled by the particle from $t = 4$ second to $t = 6$ second. [3]
- Q.2** a) Write the expression for distance travelled in n^{th} second of a body is having linear motion. [2]
- b) A ball is tossed with a velocity of 10 m/s directly vertically upward from a window located 20 m above the ground. Knowing that the acceleration of the ball is constant and equal to 9.81 m/s^2 downward. Determine:
i) the velocity v and elevation y of the ball above the ground at any time t ,
ii) the highest elevation reached by the ball and corresponding value of t ,
iii) the time when the ball will hit the ground and the corresponding velocity.
Draw the $v - t$ and $y - t$ curves. [3]
- Q.3** a) Define linear impulse and linear impulse of a particle of mass, m acted upon by a force, F . [2]
- b) A 10 kg package drops from a downhill into a 25 kg cart with a velocity of 3 m/s. Knowing that the cart is initially at rest and roll freely. Determine:
i) the final velocity of the cart,
ii) the impulse exerted by the cart on the package, [3]
- Q.4** Car A and Car B are moving on a road with different velocities is shown in Fig. 1. The velocity and tangential acceleration of car A are $v_A = 10 \text{ m/s}$ and $(a_t)_A = 5 \text{ m/s}^2$ respectively whereas velocity and acceleration of car B are given by, $v_B = 18.5 \text{ m/s}$ and $a_B = 2.0 \text{ m/s}^2$. Find: a) $v_{A/B}$ and b) $a_{A/B}$ [5]

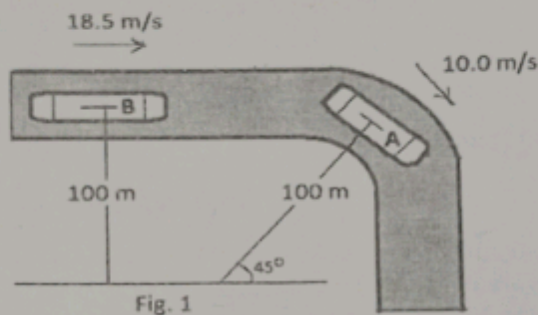


Fig. 1

Q.5

A 1000 kg car starts from rest at point 1 and moves without friction down the track shown in Fig. 2. Determine:

- the force exerted by the track on the car at point 2, where the radius of curvature of the track is 6.0 m.
- the minimum safe value of the radius of curvature at point 3.

[5]

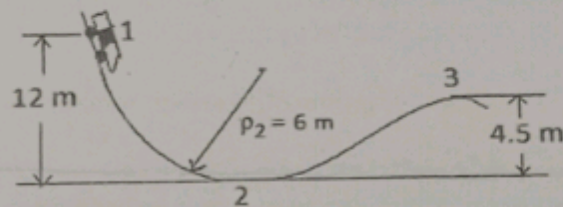


Fig. 2

Q.6

Two blocks A and B of weights 1.0 kN and 2.0 kN respectively are in equilibrium positions as shown in Fig. 3. If the coefficient of friction between the two blocks as well as the block B and the floor is 0.3, find the force, P required to move the block B.

[5]

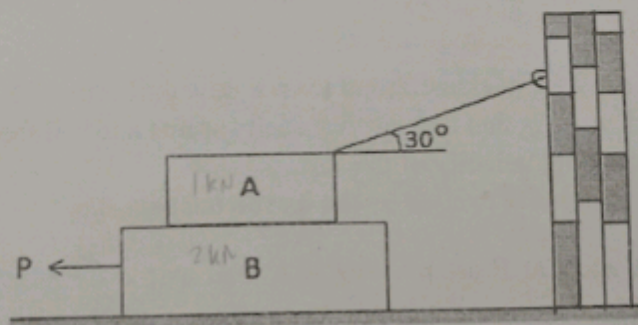


Fig. 3



B.I.T. SINDRI, DHANBAD
MECHANICAL ENGG. DEPARTMENT
SUPPLEMENTARY MID-SEMESTER EXAMINATION- 2024
B.TECH. 1ST SEMESTER

BRANCH- ME, CE, MIN, MET, CHEM & PROD. ENGG.

SUBJECT- Engg. Drawing & Computer Graphics

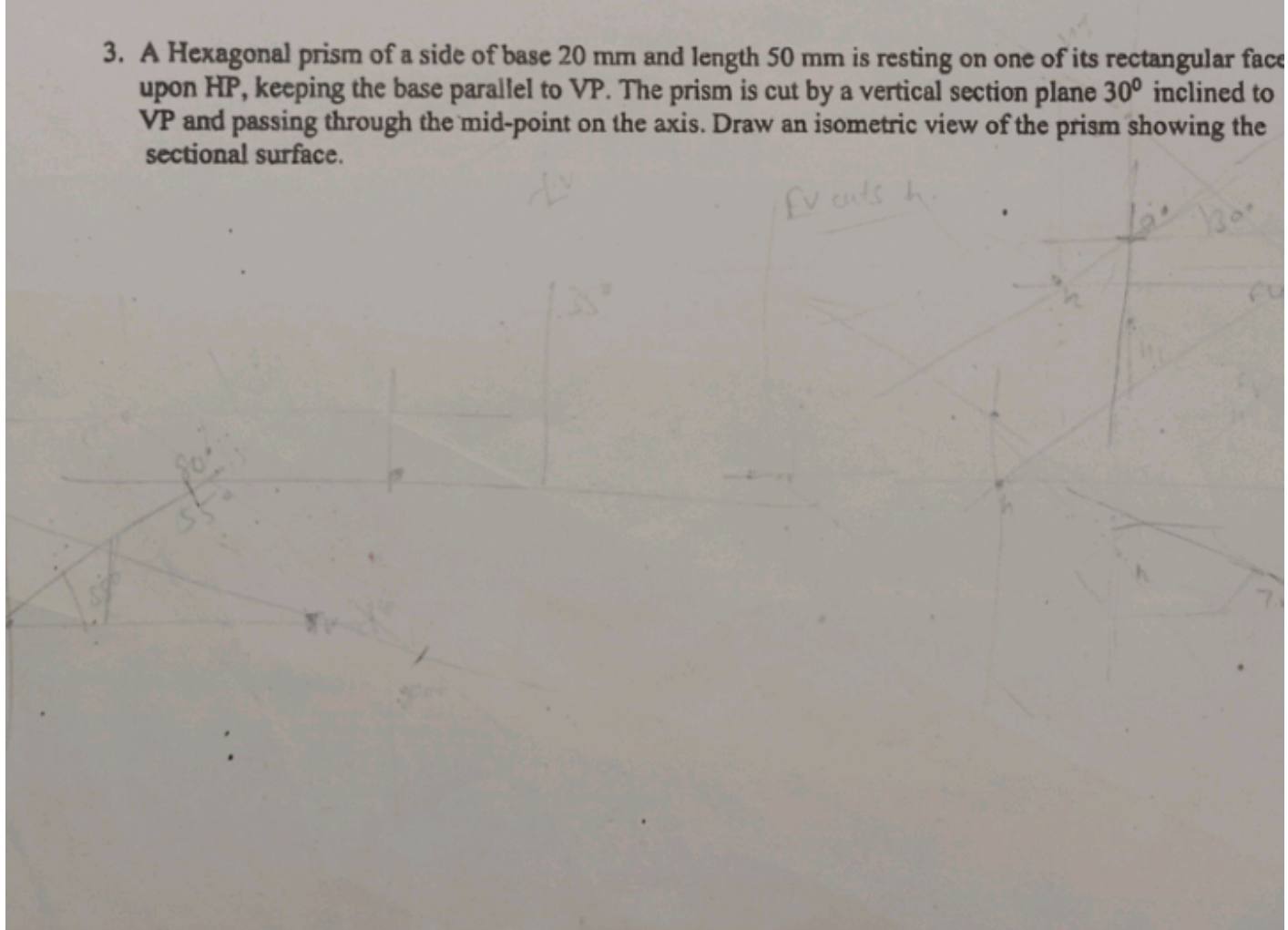
MAXIMUM MARKS-20

TIME- $1\frac{1}{2}$ Hrs.

Instructions:-

- The questions are of equal values
- Assume any missing data or values suitably.
- Draw neat and labeled Drawing
- Students can use both sides of the Drawing sheet
- Keep proper dimensioning
- Attempt any two questions

1. The Top View of a line CD measures 80 mm and makes an angle of 55° with XY. End C is in the Vertical Plane and the horizontal trace of the line is 25 mm above XY. The Line is inclined at 30° to the HP. Draw the projections of line CD. Determine its True Length, True Inclination with the VP and the VT.
2. The front view of a line PQ makes an angle 30° with XY. The HT of the line is 45 mm behind the VP. While its VT is 30 mm above the HP. The end P of the line is 10 mm below the HP and the end Q is in the First Quadrant. The line is 150 mm long. Draw the projections of the line and determine the true-length of the portion of the line which is the Second Quadrant. Also, find the angle of the line with the HP and VP.
3. A Hexagonal prism of a side of base 20 mm and length 50 mm is resting on one of its rectangular face upon HP, keeping the base parallel to VP. The prism is cut by a vertical section plane 30° inclined to VP and passing through the mid-point on the axis. Draw an isometric view of the prism showing the sectional surface.



B.TECH., FIRST SEMESTER
BRANCHES: ME, PE, ML, CL, CE AND MN
SECOND MID SEMESTER EXAMINATION, 2024
ENGINEERING DRAWING AND COMPUTER GRAPHICS (EDCG)

Full Marks: 20

Time: 1 Hour 30 Minutes

Roll No.

2	3	0			
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Important Instructions:

1. ALL THE QUESTIONS CARRY EQUAL MARKS.
2. ANSWER ANY TWO QUESTIONS.
3. The full marks corresponding to a question is indicated in right braces.
4. Assume any missing data or values suitably.
5. Write your Roll No. on the question paper in the space provided.
6. Do not write anything else on your question paper.
7. Use of a non-programmable calculator is permitted.
8. Use of any Log Tables, Slide Rules, Data Books etc. is not permitted.

1. The orthographic projections of a three-dimensional object is shown in Figure 1 below. Draw its isometric projection. [10]

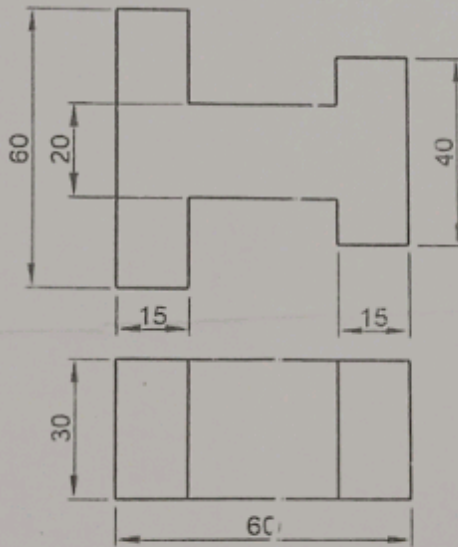


Figure 1 (Question No. 1)

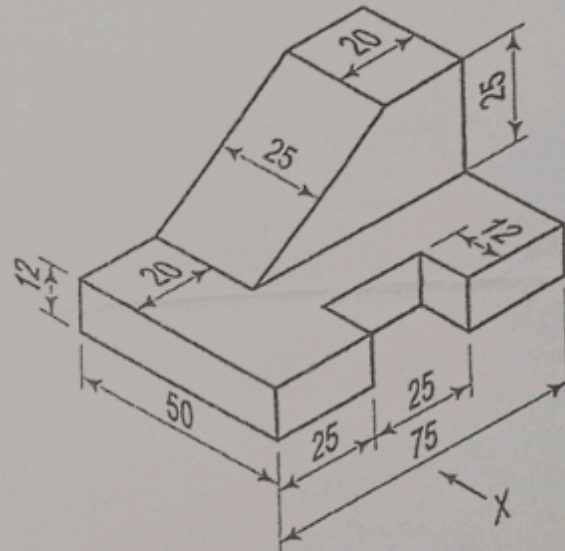


Figure 2 (Question No. 2)

All dimensions are in mm.

2. The isometric projection of a three-dimensional object is shown in Figure 2 above. Draw its orthographic projection (Plan, Elevation and Left Side View). [3+3.5+3.5=10]

3. The projectors of the ends of a line AB are 50 mm apart. The end A is 20 mm above the H.P. and 30 mm in front of the V.P. The end B is 70 mm above the H.P. and 60 mm in front of V.P.

- (i.) Draw the projections of line AB.
- (ii.) Determine the true length of line AB.
- (iii.) Determine the traces of line AB.
- (iv.) Determine the inclinations of line AB with the two planes. [2.5X4=10]

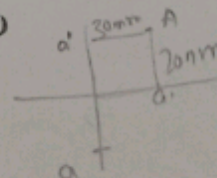
4. (i.) Differentiate between the following (Write on left bottom of drawing sheet):

(A.) First Angle Projection and Third Angle Projection [2]

(B.) Isometric Projection and Isometric View [2]

- (ii.) A cube having its side length 60 mm is placed over another cube of side length 80 mm coaxially with the vertical sides of two cubes parallel to each other. A sphere of diameter 50 mm is placed at the top first cube with its center lying on the coinciding vertical axis of two cubes. Draw its isometric projection. [6]

THE END



Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2023 (NEP)****Subject : Engineering Mechanics****Subject Code : ESEM1****Time Allowed : 3 Hours****Full Marks : 70**

*Candidates are required to give the answers in their own words as far as practicable.**Answer any five Questions in which Question No.1 is compulsory.**All questions are of equal value.*

1. Choose the correct alternative:**2×7=14**

(i) What is the dot product of two vectors which are having a magnitude equal to unity and are making an angle of 45° ?

(a) -0.707 (b) 0.707 (c) -1.414 (d) 1.414

(ii) The moment is the cross product of which two vectors?

(a) Radius and Force vectors

(b) Force and Radius vectors

(c) Radius and Force scalars

(d) Force and Radius scalars

(iii) The possible loading in various members of framed structures are

(a) compression or tension

(b) buckling or shear

(c) shear or tension

(d) all of these

(iv) Frictional force encountered after commencement of motion is called

(a) Limiting friction

(b) Kinematic friction

(c) Frictional resistance

(d) Dynamic friction

(v) The radial component of velocity for a particle moving in circular path is _____.

(a) constant

(b) radius itself

(c) variable

(d) zero

(vi) A bus travels with a speed of 15 m/s when accelerated at 0.10 m/s^2 from its rest position. What is the distance travelled?

(a) 1000 m (b) 1125 m (c) 2250 m

(d) None of these

(vii) An object is released from a certain height above the ground. Just at time it touches the ground it will possess _____.

(a) Potential energy

(b) Kinetic energy

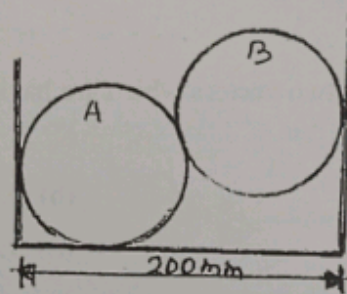
(c) Heat energy

(d) Chemical energy

2. (a) Explain moment of a force and moment of couple.

(b) Two spheres A and B of weight 1000 N and 750 N respectively are kept as shown in figure. Determine the reactions at all contact points. Radius of A is 75 mm and radius of B is 50 mm.

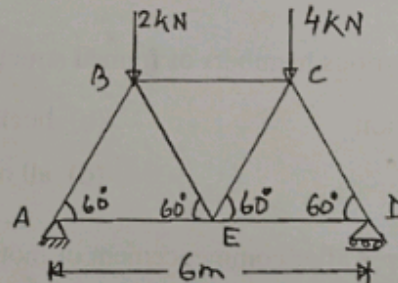
4+10



3. (a) What are the assumptions in analysis for plane truss?

(b) Find the forces in all the members of the truss shown below.

4+10



4. (a) Define coefficient of friction and limiting friction.

(b) A uniform ladder of 4 m length rests against a vertical wall with which it makes an angle of 45° . The coefficient of friction between ladder and wall is 0.4 and that between ladder and the floor is 0.5. If a man, whose weight is one-half of the ladder ascends it, how high will it be when the ladder slips?

4+10

5. (a) Derive the tangential and normal components of velocity and acceleration for a moving particle.

(b) The acceleration of a particle is given by $a = t^3 - 3t^2 + 5$, where a is in m/s^2 and t is in second. At $t = 1$ s, the displacement and velocity are 8.5 m and 6.25 m/s respectively. Calculate the displacement and velocity at $t = 2$ sec.

7+7

6. (a) Explain work, energy and power.
- (b) A hammer of mass 0.5 kg hits a nail of 25 g with a velocity of 5 m/s and drives it into a fixed wooden block by 25 mm. Find the resistance offered by the block. 6+8
7. Write short notes on *any two* of the following: 7×2
- (a) Belt friction
 - (b) Impulse momentum principle
 - (c) Equivalent system of forces
-

Jharkhand University of Technology, Ranchi

B.Tech. 1st Semester Examination, 2023 (NEP)

Subject : Basics of Electrical & Electronic Engineering

Subject Code : ESBE1

Time Allowed : 3 Hours

Full Marks : 70

Candidates are requested to give the answers in their own words as far as practicable.

The figures in the right-hand margin indicate full marks.

Answer any five questions including Question no. 1.

1. Attempt any seven questions:

2×7=14

- (i) Which of the following is an active element in a circuit?
 - (a) Current source
 - (b) Resistance
 - (c) Capacitance
 - (d) Inductance
- (ii) A $2\ \Omega$ resistance having a current of 2 Amp will dissipate the power of
 - (a) 2 W
 - (b) 4 W
 - (c) 8 W
 - (d) 10 W
- (iii) A sinusoidal current has peak value of 12 A. What is its average value?
 - (a) 7.64 A
 - (b) 24 A
 - (c) 8.84 A
 - (d) 12 A
- (iv) In pure capacitor, the voltage _____ the current by 90° .
- (v) The rating of a transformer is measured in
 - (a) Volts
 - (b) Ampere
 - (c) kW
 - (d) kVA
- (vi) The forward voltage drop across a Germanium diode is about _____.
 - (a) 2.5 V
 - (b) 3 V
 - (c) 10 V
 - (d) 0.3 V
- (vii) If the temperature of a crystal diode increases, then leakage current _____.
 - (a) remains the same
 - (b) decreases
 - (c) increases
 - (d) becomes zero
- (viii) The number of depletion layers in a transistor is _____.
 - (a) four
 - (b) three
 - (c) one
 - (d) two
- (ix) The base of a transistor is _____ doped.
 - (a) heavily
 - (b) moderately
 - (c) lightly
 - (d) None of these

- (x) The universal gate is _____.
 (a) NAND gate (b) OR gate
 (c) AND gate (d) None of these

2. (a) State Kirchhoff's Current law.
 (b) Find the mesh current i_1 , i_2 , i_3 in the circuit shown in fig. 1 by performing mesh analysis.

4+10

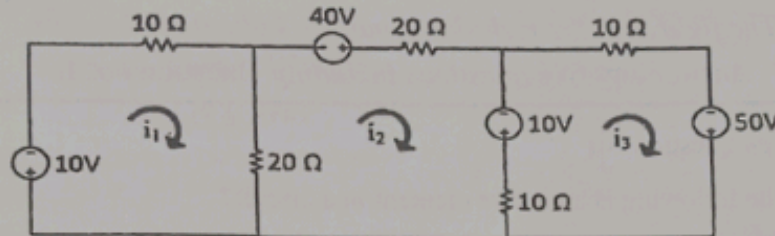


Fig. 1

3. (a) Explain the superposition theorem for ac circuits.
 (b) What is the condition of resonance in a series RLC circuit?
 (c) A series RLC circuit consists of $R = 2 \Omega$, $L = 1 \text{ mH}$ and $C = 0.2 \mu\text{F}$. Find the resonance frequency. 4+4+6
4. (a) Explain the construction details of a transformer.
 (b) How eddy current losses are minimized?
 (c) A 2300/230 V, 50 Hz, single phase transformer rating is 25 kVA. It has the following resistances and leakage reactances: $R_1 = 0.6 \Omega$ and $X_1 = 3.0 \Omega$; $R_2 = 0.008 \Omega$ and $X_2 = 0.02 \Omega$. Calculate the equivalent resistance and equivalent reactance referred to the secondary side. 4+2+8
5. (a) Discuss the operation of a $p-n$ junction diode.
 (b) In a Zener regulated power supply a Zener diode with $V_z = 6 \text{ V}$ is used for regulation. The load current is to be 4 mA and the unregulated input is 10 V. What should be the value of series resistor R? 6+8
6. (a) Simplify the Boolean function $F(x, y, z) = \sum(0, 2, 4, 5, 6)$, using K-map method.
 (b) Write the logical expression of half adder with the help of its truth table and draw its circuit diagram using logic gates. 7+7
7. Write short notes on *any two* of the following: 7×2=14
 (a) Full wave rectifier
 (b) Hysteresis loop
 (c) X-OR gate

Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2023 (NEP)****Subject : Engineering Mathematics-I****Subject Code : BSM01****Time Allowed : 3 Hours****Full Marks : 70**

*Candidates are requested to give the answers in their own words as far as practicable.**The figures in the margin indicated full marks.**Answer any five questions. Question No.1 is compulsory.*

1. Answer the following questions:**2×7=14**(i) If $A = PDP^{-1}$ then A^n equals to

(a) $P^n D (P^{-1})^n$

(b) $P^n D P^{-1}$

(c) $PD(P^{-1})^n$

(d) $PD^n P^{-1}$

(ii) Let V be a vector space of all real numbers R over R . Which of the following is a vector subspace of V ?

(a) $C(R)$

(b) $R(R)$

(c) $Q(R)$

(d) $Z(R)$

(iii) If $u + v = x$, $uv = y$, then $\frac{\partial(x,y)}{\partial(u,v)}$ is equal to

(a) $u - v$

(b) uv

(c) $u + v$

(d) None of these

(iv) The necessary condition that the point (a, b) is a _____ point of $f(x, y)$ if $f_x(a, b) = f_y(a, b) = 0$.

(a) maximum

(b) stationary

(c) saddle point

(d) minimum

(v) The value of $\int_0^\infty e^{-x} x^{\frac{3}{2}} dx$ is

(a) $\frac{3}{4}\sqrt{\pi}$

(b) $\frac{5}{4}\sqrt{\pi}$

(c) $\frac{3}{5}\sqrt{\pi}$

(d) $\frac{1}{4}\sqrt{\pi}$

(vi) The value of $\int_3^6 \int_2^4 \int_1^2 dx dy dz$ is

(a) 2

(b) 4

(c) 6

(d) 10

- (vii) Let $\vec{F}$ be the force acting on a particle, then $\int_C \vec{F} \cdot d\vec{r}$ represents _____ by the particle.
- (a) Work done (b) Circulation
(c) Conservative field (d) None of these
2. (a) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 7 & -1 & 3 \\ 6 & 1 & 4 \\ 2 & 4 & 8 \end{bmatrix}$ and hence find its inverse.
- (b) Show that the equations $3x + 4y + 5z = a$; $4x + 5y + 6z = b$; $5x + 6y + 7z = c$ do not have a solution unless $a + c = 2b$. 7+7
3. (a) If $T: R^2 \rightarrow R^3$ is a linear transform defined by $T(x_1, x_2) = (x_1 - x_2, x_2 - x_1, -x_1)$. Find a basis and dimension for $R(T)$ and $N(T)$. Also, verify Rank-Nullity theorem.
- (b) If $x^x y^y z^z = c$, then show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$. 7+7
4. (a) The temperature T at any point (x, y, z) in space is $T(x, y, z) = kxyz^2$. Find the highest temperature on the surface of the sphere $x^2 + y^2 + z^2 = a^2$.
- (b) If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$, where $x = r \cos \theta$, $y = r \sin \theta$, show that $\frac{\partial(u,v)}{\partial(r,\theta)} = 6r^3 \sin 2\theta$. 7+7
5. (a) Prove that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^2}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}$.
- (b) Change the order of integration and hence evaluate the integral $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x dx dy}{\sqrt{x^2+y^2}}$. 7+7
6. (a) Apply double integration to calculate the volume generated by the revolution of the cardioid $r = a(1 + \cos \theta)$ about its axis.
- (b) Find the value of a so that the vector $\vec{F} = (ax^2y + yz)\vec{i} + (xy^2 - xz^2)\vec{j} + (2xyz - 2x^2y^2)\vec{k}$ is solenoidal. Also, find the curl of the vector $\vec{F}$. 7+7
7. (a) If $\vec{A} = (3x^2 + 6y)\vec{i} - 14yz\vec{j} + 20xz^2\vec{k}$, then evaluate $\int \vec{A} \cdot d\vec{R}$ from $(0,0,0)$ to $(1,1,1)$ along the path $x = t, y = t^2, z = t^3$.
- (b) Verify Green's theorem for $\int_C [(x^2 - \cosh y)dx + (y + \sin x)dy]$, where C is the rectangle with vertices $(0,0), (\pi, 0), (\pi, 1), (0,1)$. 7+7

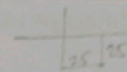
Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2023 (NEP)****Subject : Engineering Chemistry****Subject Code : BSC02****Time Allowed : 3 Hours****Full Marks : 70**

*Candidates are requested to give the answers in their own words as far as practicable.**Answer any five questions in which Question No. 1 is compulsory.**All questions are of equal value.*

1. Answer the following questions :**2×7=14**

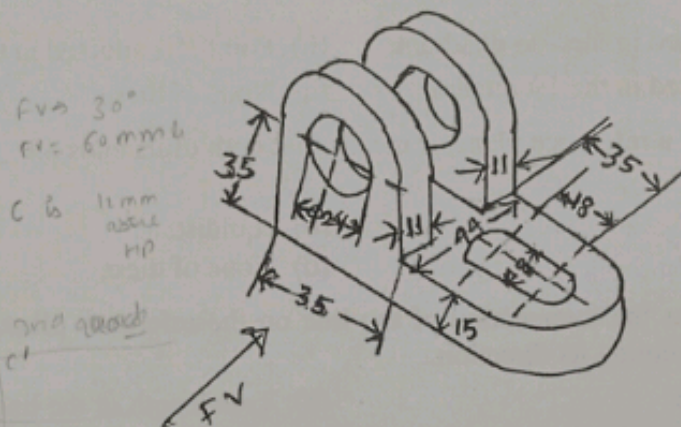
- (i) Which of the following wavelength fall in IR spectrum?
- (a) 2000 nm (b) 110 nm
(c) 800 nm (d) 10 cm
- (ii) Identification of an organic compound that can established by using
- (a) IR spectroscopy (b) UV spectroscopy
(c) Any of these (d) None of these
- (iii) The metal at the top of electrochemical series is
- (a) More active (b) Less active
(c) Most stable (d) More noble
- (iv) Difficult to monitor and very dangerous form of corrosion—
- (a) Pitting (b) Galvanic
(c) Crevice (d) Stress
- (v) The nanomaterials have the following characteristics—
Which do not exist in the corresponding bulk materials?
- (a) High surface energy (b) Greater imperfection
(c) Smaller number of surface atom (d) None of these
- (vi) Which of the following nanomaterials is utilized as a solid-fuel in rocket propulsion?
- (a) Nano aluminium (b) Nano zirconium
(c) Nano silicate (d) All of these
- (vii) The capacity of a cell is measured in
- (a) Ampere - hour (b) Amperes
(c) Watts (d) Watt - hours

2. (a) Discuss about following:
- (i) Quality control in engineering context
 - (ii) Role of refractories in engineering
- (b) Write the principle of UV spectroscopy. Discuss various electronic transitions. (3.5×2)+7
3. (a) State and explain electrochemical theory of corrosion.
- (b) Discuss about various corrosion prevention techniques. 7+7
4. (a) Define battery. Differentiate between electrolytic cell and electrochemical cell.
- (b) Explain principle, working and application of fuel cells. 7+7
5. (a) Define nanomaterials. Explain the mechanical and magnetic properties of nanomaterials.
- (b) Explain various importance of nanomaterials. 7+7
6. (a) Explain top-down and bottom-up approach of nanomaterials.
- (b) Discuss about green hydrogen technology. 7+7
7. Write notes on following: 3.5×4
- (a) chromophores
 - (b) auxochromes
 - (c) HPLC
 - (d) Galvanic corrosion

Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2023 (NEP)****Subject : Engineering Drawing & Computer Graphics****Subject Code : ESED2****Time Allowed : 3 Hours****Full Marks : 70***Candidates are requested to give the answers in their own words as far as practicable.**Answer any five questions in which Question No. 1 is compulsory.**All questions are of equal value.**Assume any missing data or values suitably.***1. Answer the following questions :****2×7=14**

- (i) A point P is 25 mm below the HP and 25 mm in front of the VP. State the quadrant in which the point is situated.
- (a) Point P is situated in the 2nd quadrant. (b) Point P is situated in the 4th quadrant.
 (c) Point P is situated in the 1st quadrant. (d) None of these
- (ii) If a line is parallel to a reference plane, it means that both of its ends are _____ from that plane.
- (a) non-equidistant (b) equidistant
 (c) Both (a) and (b) (d) None of these
- (iii) _____ is defined as the intersection of the line on the reference plane when it is moved towards it without changing its direction.
- (a) Trace of the line (b) True length of the line
 (c) Apparent length of the line (d) None of these
- (iv) When a line is inclined to both VP and HP but parallel to PP (Profile Plane) then sum of its inclination will be
- (a) $\theta + \phi = 45^\circ$ (b) $\theta + \phi = 90^\circ$
 (c) $\alpha + \beta = 45^\circ$ (d) $\theta + \phi = 120^\circ$
- (v) Isometric projection of a sphere is draw with true (length) radius. But its centre is located with _____.
- (a) isometric height (b) true height
 (c) half of the true height (d) half of the isometric height
- (vi) In an orthographic projection approaches, if a face is perpendicular to the direction of viewing, its _____ will be seen in that view.
- (a) tilted shape and size (b) apparent shape and size
 (c) true shape and size (d) comparatively smaller shape and size
- (vii) If a line is inclined to the HP and VP then it will have _____.
- (a) both HT and VT (b) only HT
 (c) only VT (d) no traces in this case

2. A line AB, 50 mm long, has its end A away from the HP and the VP than end B. The line is inclined to the HP at 30° and to the VP at 45° . Draw the projections if end A is 35 mm above the HP and 50 mm in front of the VP. 14
3. Line CD is in the second quadrant and has 25° inclination with HP, while the front view has 30° inclination with xy line and 60 mm length. If the end C is 12 mm above HP and the end D is 60 mm behind VP, draw its projections. 14
4. A cylindrical block of base, 60 mm diameter and height 90 mm, standing on the HP with its axis perpendicular to the HP. Draw its isometric view. 14
5. A pentagonal pyramid of base side 30 mm and axis length 65 mm is resting on HP on its base with a side of base perpendicular to VP. It is cut by a plane inclined at 30° to HP and perpendicular to VP and passes through a point at a distance 30 mm from the apex. Draw the Isometric projection of the truncated pyramid. 14
6. The pictorial view of bearing block is shown below. Draw the front view, top view and side view according to the 1st angle projection method. (All dimensions are in mm.) 14



7. The pictorial view of a block is shown below (Fig. is not to scale). Draw the top view, front view and side view according to the first angle projection method. All dimensions are in mm. 14

